

Breaking the low sample rate equals low power paradigm using an event based vision approach for hand tracking in AR

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ABSTRACT

As spatial computers move towards glasses, where constraints on size, weight and power are critical, a shift in how we use sensors is needed. Ultraleap has developed the first hand tracking pipeline using event cameras on an AR headset. Event cameras allow to break the frame-imaging paradigm of "low power equates to low sample rate" and enhance user experience while maintaining low power budget.

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1 INTRODUCTION

The adoption of spatial computing is going to be driven by a number of factors. One transformation that is needed is to move away from an indoor device towards an all day, everywhere wearable. Glasses represent a natural form-factor choice, but require solving some fundamental challenges with tight constraints: size, weight and power. At Ultraleap we are focused on user input using natural hand-based user interactions (UI). Bringing that capability to a glasses form factor requires rethinking the sensing paradigm. In the frame-based approach the world is sampled at a fixed rate, which dictates power: the lower the sample rate, the lower the power. This is in contrast to what is required to ensure a good experience. Switching to an event-based platform breaks this tension: an event sensor has a biologically inspired architecture, responding to changes in light in the scene asynchronously. This allows a high sample rate of the world to be achieved whilst maintaining a low power budget. The system presented by Ultraleap is the first hand tracking pipeline using event cameras and, to our knowledge, the first to

be demonstrated publicly on a HMD, with previous works using static cameras [Rudnev et al. 2021]. Our imaging system enables the skeletal 3D points that represent the user's hand to be updated at 240fps while the cameras consume approximately 20mW. By comparison, a frame-based camera running at 120fps consume typically 150-200mW. The update frequency is no longer limited by the sampling of the camera: the two have been decoupled, allowing for dynamic event driven update rates in the future. This is the first step to exploiting event-based technology to solve the problems of a spatial computing perception stack that is both highly capable and works within the power budget of glasses.

2 NORTH STAR AR HEADSET

The Leap Motion North Star [Leap Motion 2018] headset uses a curved optical combiner to produce a virtual image in front of the user. One of the most impressive elements of this headset is its field of view, which is close to 100° . A pair of Prophesee GenX320 [Prophesee 2023] event sensors are positioned on the headset, at a baseline of 150mm. The sensors have ultra-wide angle lenses, reaching 170° horizontal and vertical FOV. A near-IR LED (850nm) is placed at the center of the baseline. The GenX320 sensors are 320×320 with $6.3\mu\text{m}$ pixels. They require very low power, approximately 20mW, and have high dynamic range, a key feature for a device operating in a wide number of illumination environments. The first assembled prototype can be seen in Figure 1.

The event streams are synchronised and fed to a tracking pipeline that includes event signal processing and inference using Ultraleap's hand tracking ML model. The output from this is made available to the application layer. Our demo uses a Monado OpenXR runtime, and the application was developed using StereoKit [StereoKit 2023].

3 VISITOR EXPERIENCE

The experience is built around a paint application developed: the user pinches their right hand to initialise the brush. Moving the hand produces a visible trace, as shown in Figure ???. The user pick different colours by making a fist with their left hand, which prompts palette to appear on an anchored menu. The hand overlay can be switched on/off to enable to cue more of their real hand and less on the virtual hands to allow for a more AR like experience.

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Figure 1: First iteration of the event-sensor based HMD at Ultraleap.



Figure 2: The demo app, showing the user's hand drawing in the air.

4 CONCLUSION

Spatial computing is a growing discipline, with exciting challenges to tackle as we strive towards a glasses form factor. The most natural way to interact in this new computing paradigm is using our hands. Ultraleap are pushing the boundaries of hand tracking, exploring the latest sensing technology to show people what could be possible in the future.

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